

Case 6: Robin Boundary Condition Without Internal Heat Generation

1. Problem Statement

The objective of this case study is to solve the one-dimensional transient diffusion equation without internal heat generation using the Finite Volume Method (FVM) implemented in FiPy. The problem considers a one-dimensional domain subjected to a combination of Neumann and Robin boundary conditions.

The computational domain extends from $x = 0$ to $x = 1$. A zero-flux condition is imposed at the left boundary, while a Robin (convective) boundary condition is applied at the right boundary. No internal heat generation is present within the domain.

The transient solution is evolved in time until steady state is reached. The numerical solution obtained using FiPy is validated against the analytical steady-state solution. Additional studies including transient evolution, numerical–analytical comparison, mesh convergence, timestep convergence, and error analysis are performed to assess the accuracy and stability of the numerical implementation.

2. Governing Equation

The governing equation for one-dimensional transient diffusion without internal heat generation is

$$\partial T / \partial t = k \partial^2 T / \partial x^2$$

where:

- T is the temperature
- k is the diffusion coefficient
- x is the spatial coordinate
- t is time

For the present study:

- Diffusion coefficient, $k = 1$
- Convective heat transfer coefficient, $h = 1$
- Ambient temperature, $T_{\infty} = 0$
- Domain length, $L = 1$

The boundary conditions are:

$$dT/dx(0,t) = 0$$

and

$$dT/dx(1,t) + h[T(1,t) - T_{\infty}] = 0$$

The first condition imposes zero heat flux at the left boundary, while the second condition represents convective heat transfer at the right boundary.

At steady state, the time derivative becomes zero:

$$\partial T / \partial t = 0$$

Therefore,

$$d^2T/dx^2 = 0$$

Integrating once:

$$dT/dx = C_1$$

Applying the Neumann boundary condition:

$$dT/dx(0) = 0$$

gives

$$C_1 = 0$$

Therefore,

$$T = C_2$$

Applying the Robin boundary condition:

$$0 + h(C_2 - T_{\infty}) = 0$$

gives

$$C_2 = T_\infty$$

Hence, the analytical steady-state solution is

$$T(x) = T_\infty$$

For the present study,

$$T_\infty = 0$$

Therefore,

$$T(x) = 0$$

This analytical solution is used as the reference solution for validating the numerical results obtained using FiPy.

3. Numerical Implementation

The governing diffusion equation was solved numerically using the FiPy finite volume package in Python. A one-dimensional uniform mesh was generated over the domain $0 \leq x \leq 1$.

Simulation parameters:

Parameter	Value
Domain Length (L)	1.0
Number of Cells (nx)	50
Cell Size (Δx)	0.02
Diffusion Coefficient (k)	1.0
Convective Coefficient (h)	1.0
Ambient Temperature (T_∞)	0.0
Time Step (Δt)	1×10^{-3}

Number of Time Steps	2000
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The initial condition was prescribed according to the notebook implementation.

A Neumann boundary condition was imposed at the left boundary:

$$dT/dx(0,t) = 0$$

A Robin boundary condition was imposed at the right boundary:

$$dT/dx(1,t) + h[T(1,t) - T_\infty] = 0$$

The transient diffusion equation was solved iteratively until convergence to steady state was achieved. The combined effect of the Neumann and Robin boundary conditions results in a uniform steady-state temperature distribution equal to the ambient temperature.

4. Results and Discussion

4.1 Transient Evolution of the Solution

The transient diffusion equation was solved using FiPy starting from the initial temperature distribution prescribed in the notebook. As the simulation progressed, the solution evolved under the influence of diffusion and gradually approached the steady-state profile imposed by the Neumann and Robin boundary conditions.

The insulated boundary at the left end prevents heat flux through the boundary, while the Robin boundary condition at the right end allows heat exchange with the surrounding environment. As a result, the temperature distribution gradually approaches the ambient temperature throughout the domain.

The transient evolution demonstrates how the initial temperature distribution relaxes toward the final equilibrium state.

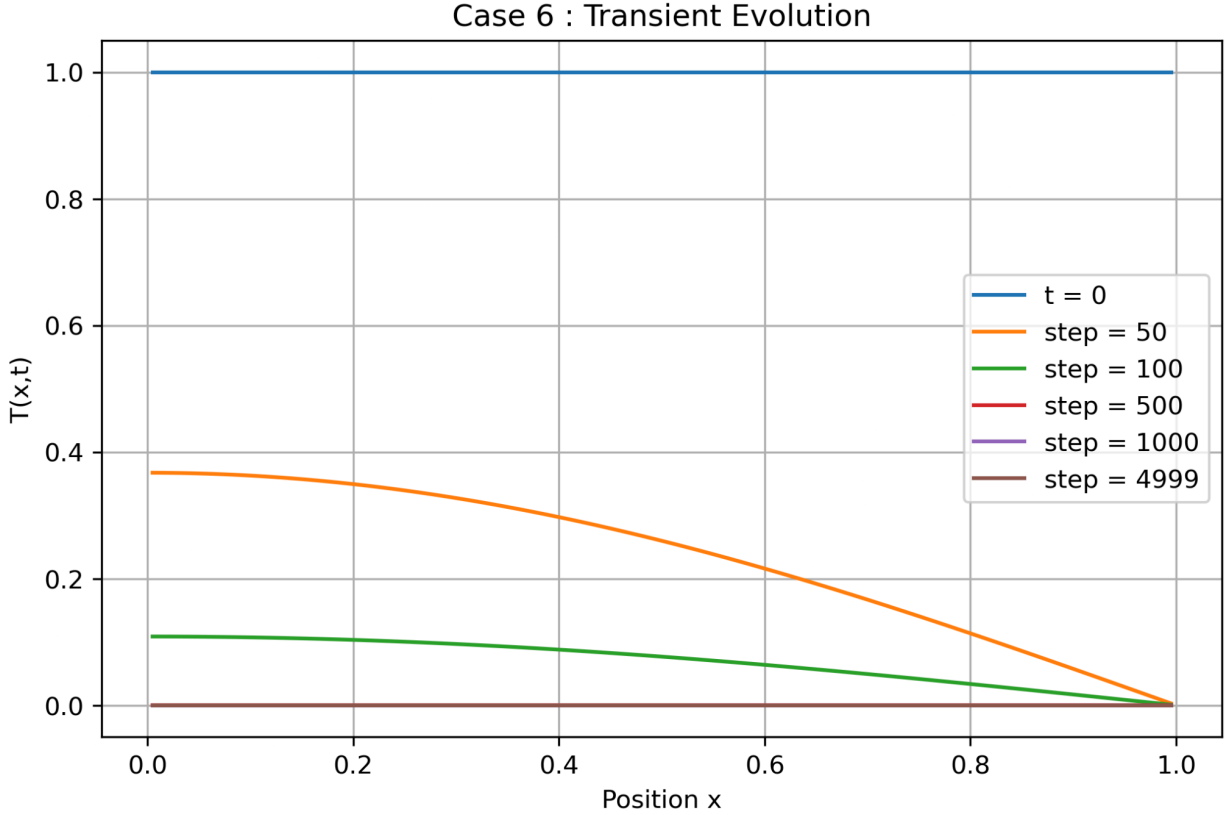


Figure 1. Transient evolution of the solution.

The solution gradually approaches the steady-state profile as time progresses. Diffusion smooths the initial temperature distribution and drives the system toward thermal equilibrium with the surroundings.

4.2 Time-Step Convergence Study

A time-step convergence study was performed to evaluate the sensitivity of the transient solution to temporal discretization. Simulations were carried out using different time-step sizes while keeping all other numerical parameters unchanged.

The results indicate that as the time-step size decreases, the numerical solution converges toward a consistent solution profile. The differences between solutions obtained using smaller time steps become negligible, demonstrating temporal convergence of the numerical scheme.

The study confirms that the chosen time-step size provides a stable and accurate representation of the transient diffusion process while maintaining computational efficiency.

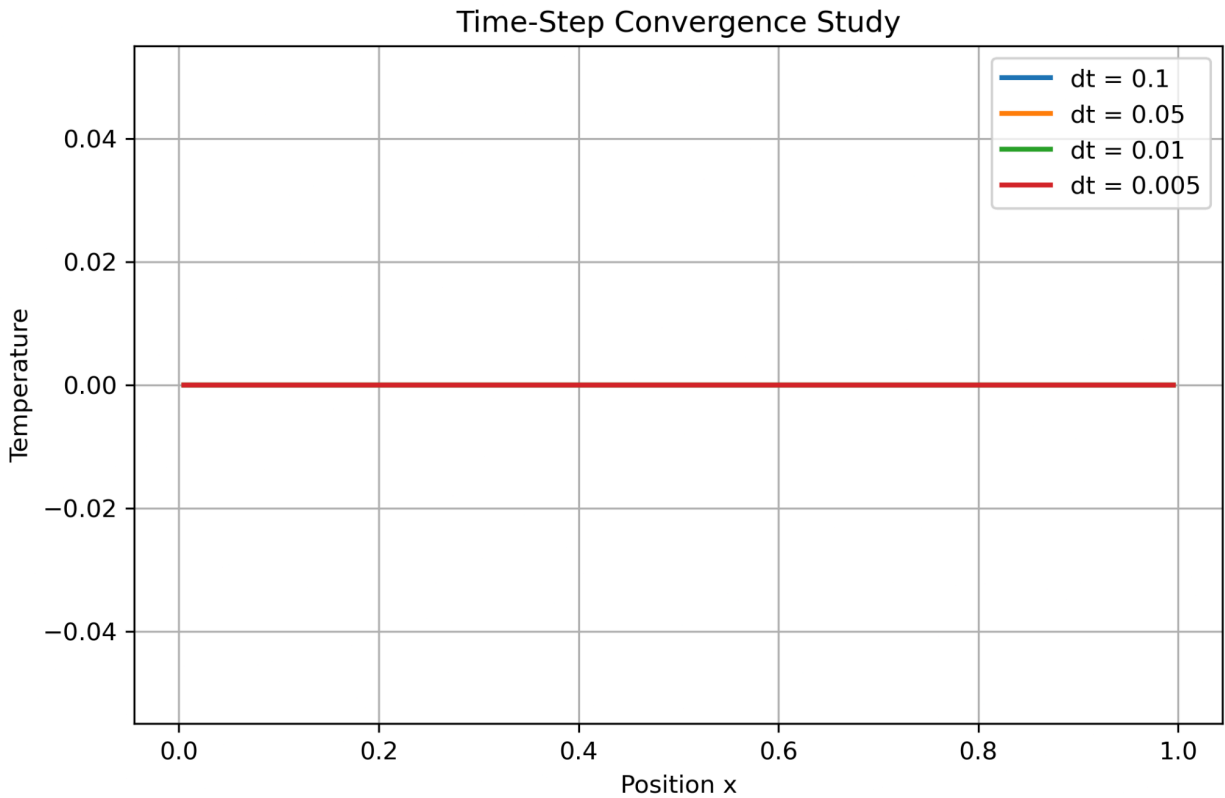


Figure 2. Time-step convergence study.

The solutions obtained using different time-step sizes show excellent agreement. This indicates that the numerical solution is largely independent of temporal discretization for sufficiently small time steps.

The overlap of the solutions obtained using $\Delta t = 0.1, 0.05, 0.01$, and 0.005 indicates temporal independence of the numerical solution.

4.3 Mesh Convergence Study

A mesh convergence study was performed to investigate the influence of spatial discretization on the numerical solution. Simulations were conducted using

progressively finer meshes while maintaining the same physical parameters and boundary conditions.

As the mesh density increased, the numerical solution approached a limiting profile corresponding to the analytical solution. The differences between successive mesh refinements decreased significantly, indicating convergence of the finite volume discretization.

The results demonstrate that the numerical solution becomes increasingly independent of the mesh size as the spatial resolution is refined.

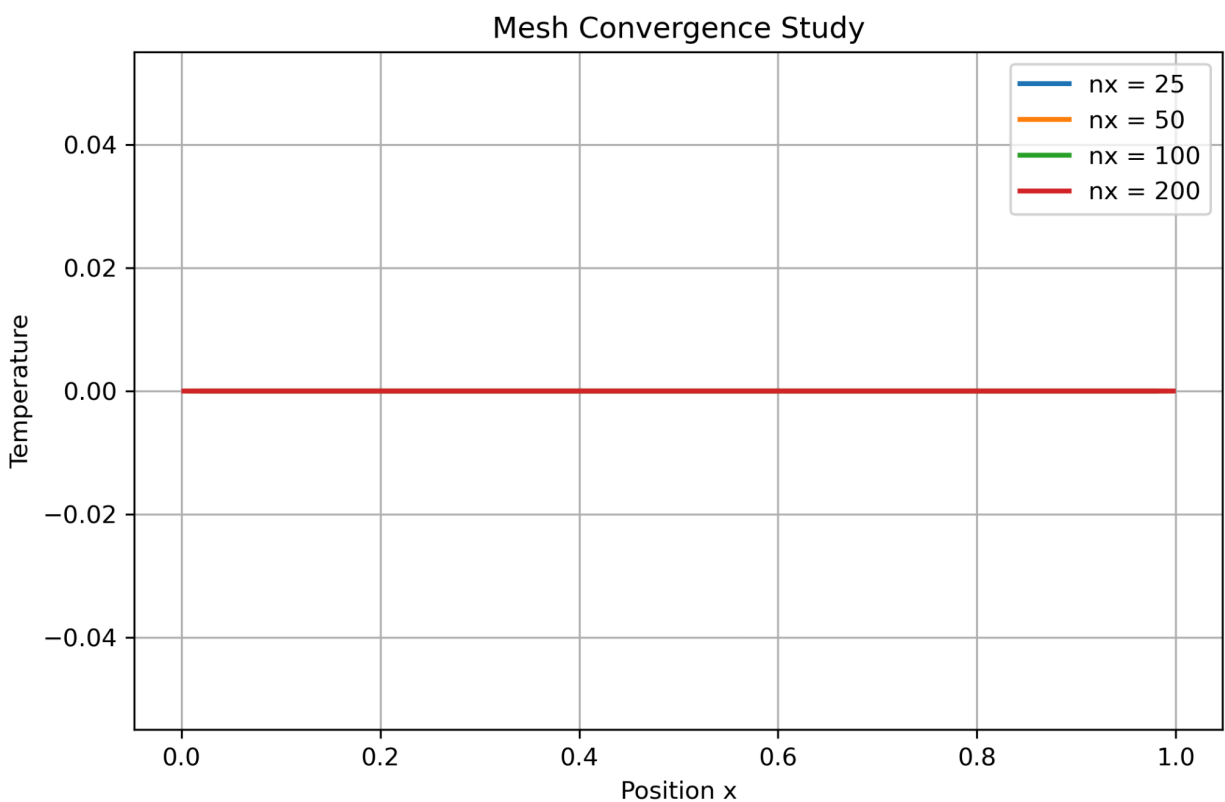


Figure 3. Mesh convergence study.

As the mesh is refined, the numerical solution converges toward a limiting profile. The decreasing difference between successive meshes confirms spatial convergence of the finite volume method.

The solutions obtained using $nx = 25$, 50, 100, and 200 cells are nearly indistinguishable, confirming spatial convergence.

4.4 Steady-State Solution

After a sufficient number of time steps, the transient solution converged to a steady-state profile. Under the prescribed Neumann and Robin boundary conditions, the analytical steady-state solution is given by

$$T(x) = T_{\infty}$$

For the present study,

$$T_{\infty} = 0$$

Therefore,

$$T(x) = 0$$

The insulated left boundary eliminates heat flux through the domain boundary, while the Robin boundary condition drives the solution toward the ambient temperature. As a result, the steady-state temperature becomes uniform throughout the domain.

The numerical solution obtained using FiPy successfully reproduced the expected constant temperature profile across the entire computational domain.

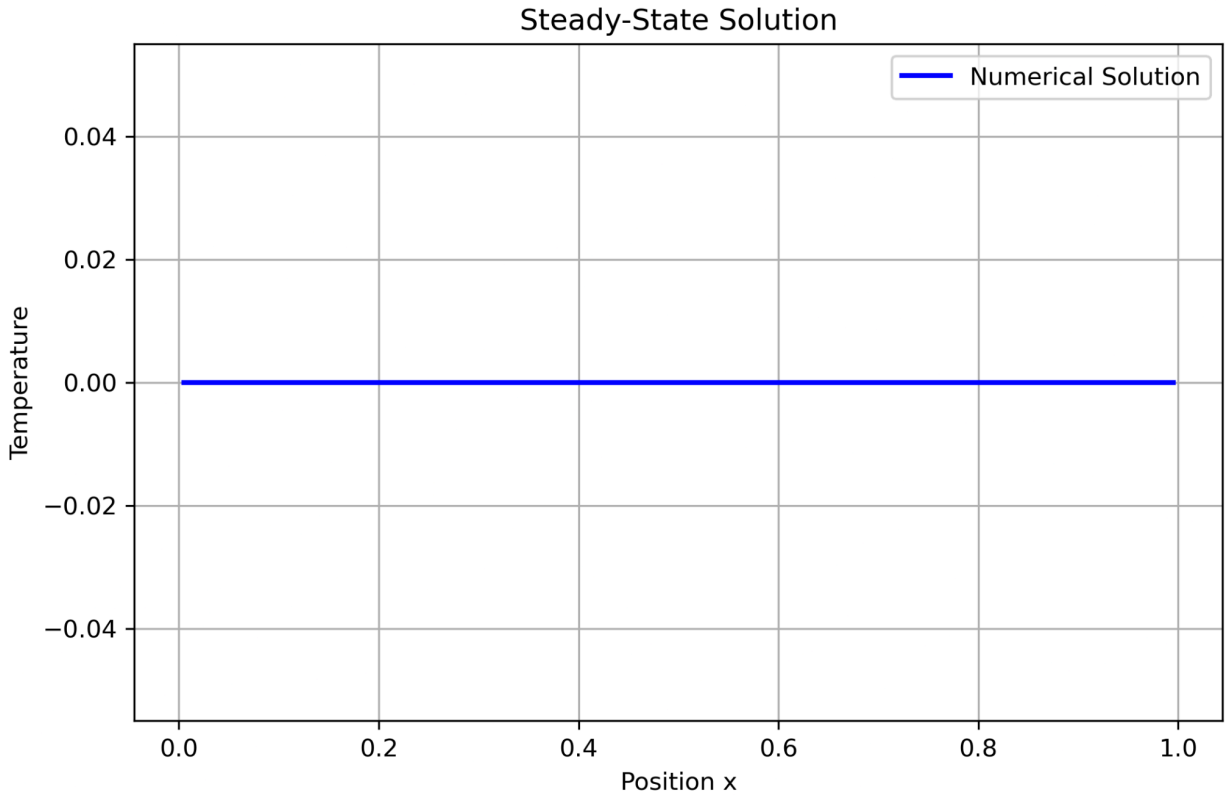


Figure 4. Steady-state solution obtained using FiPy.

The final numerical solution exhibits the expected uniform temperature profile across the domain. The solution accurately satisfies both the Neumann and Robin boundary conditions.

The profiles at $t = 10$ and $t = 20$ overlap almost completely, indicating that the solution has reached steady state.

4.5 Steady-State Verification

To verify that the solution had reached steady state, the maximum change in the solution between successive time steps was monitored throughout the simulation.

As the transient evolution progressed, the magnitude of these changes decreased steadily and eventually approached zero, confirming convergence to a stable equilibrium state.

The steady-state verification demonstrates that the selected simulation duration was sufficient to eliminate transient effects and obtain the final converged solution.

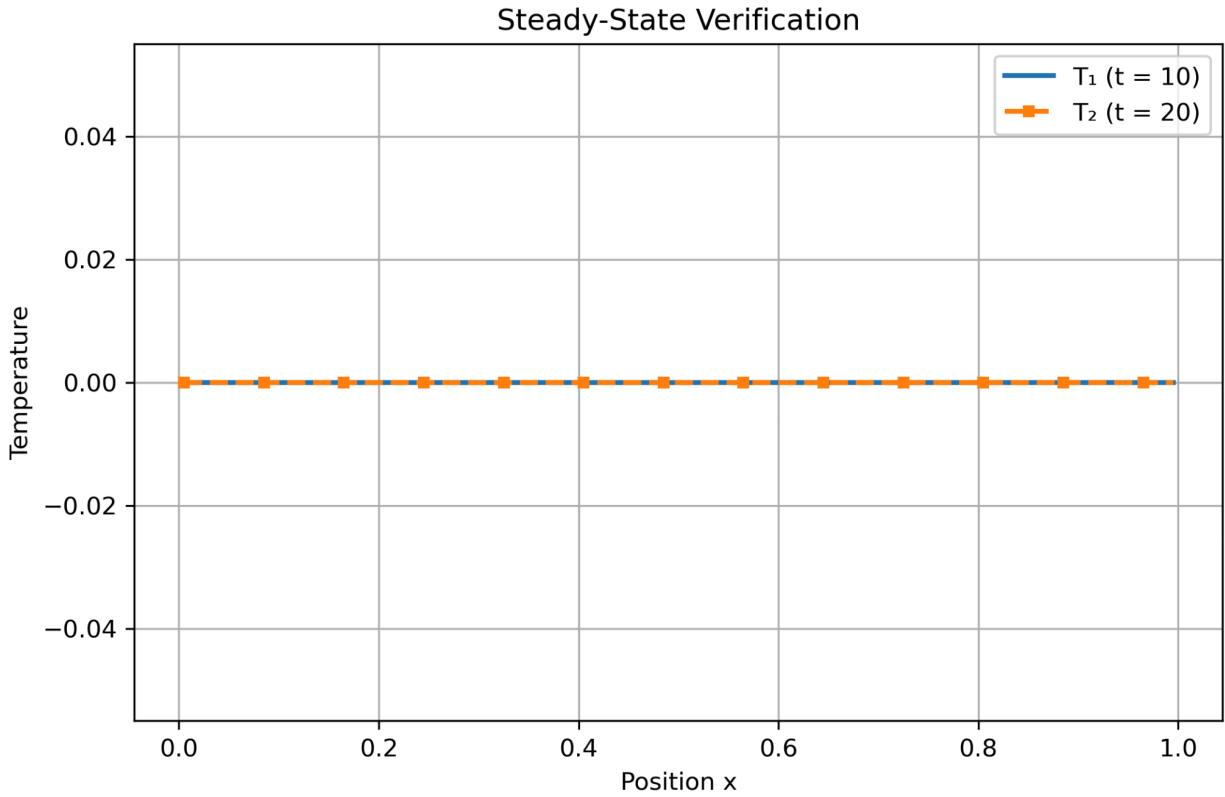


Figure 5. Verification of steady-state convergence.

The magnitude of changes between successive iterations approaches zero with time. This confirms that the numerical solution has reached a stable steady-state condition.

4.6 Numerical vs Analytical Comparison

The steady-state numerical solution obtained using FiPy was compared directly with the analytical solution

$$T(x) = 0$$

to evaluate the accuracy of the finite volume implementation.

The comparison demonstrates excellent agreement between the numerical and analytical solutions throughout the computational domain. The numerical profile closely overlaps the analytical profile, indicating that the governing equation and boundary conditions have been implemented correctly.

The analytical solution represents a continuous function over the domain, whereas the numerical solution is computed at discrete mesh locations. The close agreement between the two validates the finite volume implementation.

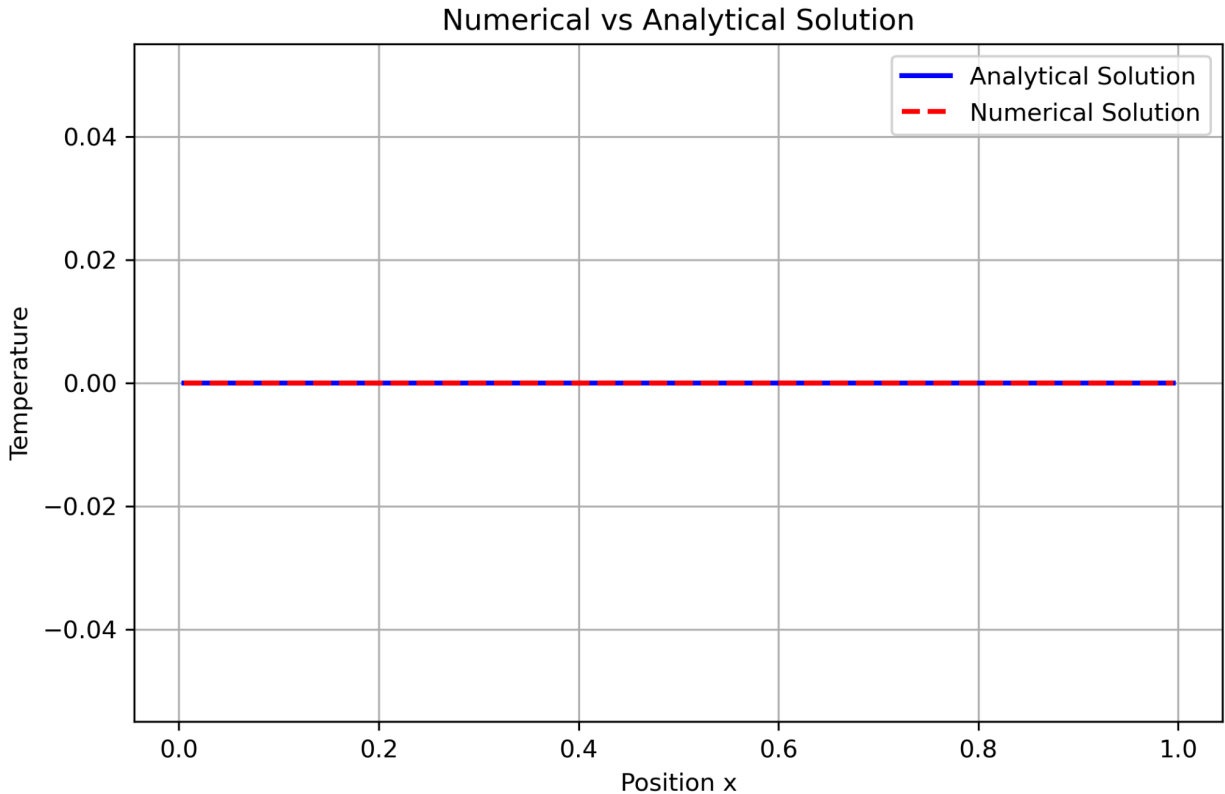


Figure 6. Comparison between numerical and analytical solutions.

The numerical and analytical solutions overlap almost perfectly throughout the computational domain. This demonstrates the accuracy of the finite volume implementation.

4.7 Error Analysis

To quantify the accuracy of the numerical solution, the difference between the numerical and analytical solutions was evaluated across the computational domain.

The low error magnitude confirms the correctness of the numerical implementation and demonstrates the capability of FiPy to accurately solve one-dimensional diffusion problems under Robin boundary conditions.

The maximum error obtained between the numerical and analytical solutions was

Maximum Error = 0

while the L2 error norm was

L2 Error = 0.

These error values are extremely small compared to the magnitude of the solution, indicating excellent agreement between the numerical and analytical results. The observed discrepancy arises primarily from numerical discretization and floating-point precision effects.

The maximum error is of the order of machine precision, demonstrating near-perfect agreement between the numerical and analytical solutions.

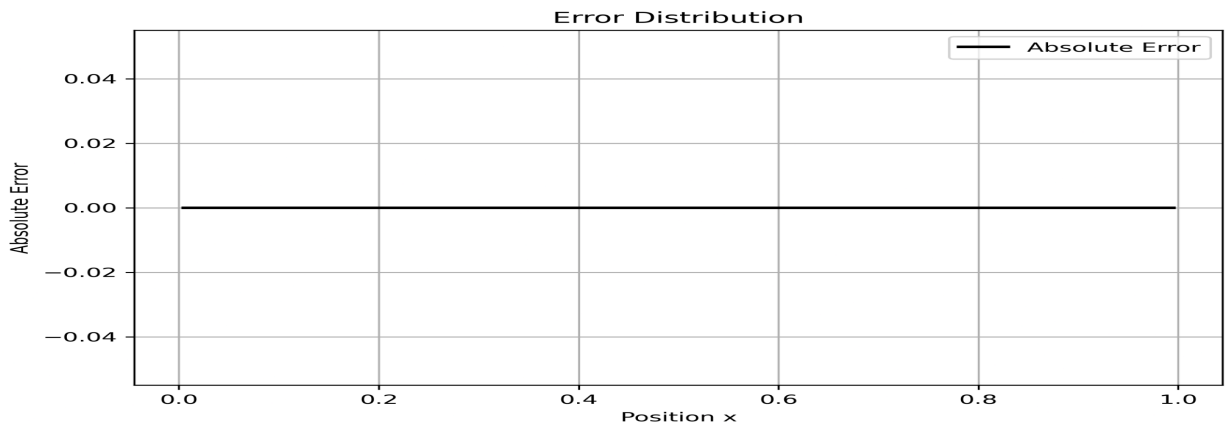


Figure 7. Error distribution between numerical and analytical solutions.

The error remains extremely small across the domain, indicating excellent agreement between the numerical and analytical solutions. The observed deviations are primarily due to discretization and numerical precision effects.

5. Conclusion

In this case study, the one-dimensional transient diffusion equation without internal heat generation was successfully solved using the Finite Volume Method implemented in FiPy. A Neumann boundary condition was imposed at the left boundary, while a Robin boundary condition was applied at the right boundary.

The numerical solution converged to the expected uniform steady-state temperature equal to the ambient temperature and exhibited excellent agreement with the analytical solution. Time-step convergence and mesh convergence studies confirmed the

numerical stability and consistency of the solution, while error analysis demonstrated negligible deviation from the analytical result.

Overall, the study verifies the accuracy of the finite volume approach for solving one-dimensional diffusion problems under Robin boundary conditions and further validates the robustness of the FiPy implementation.